Alberto Estrada Gómez

AI & Machine Learning Engineer

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Computer engineer focused on applied AI: I build forecasting systems end to end and evaluate them rigorously. I currently bring AI and automation to real business processes at MicroCAD.

EXPERIENCE

AI & Process Automation Engineer · MicroCAD Informática — Sep 2026 – Present

- Early member of the new AI & Automation department: first solutions shipped in **under 2 months**.
- Automate processes with **Power Automate** and Python; integrate external APIs with **SharePoint** and in-house apps.
- Build internal apps and **Power BI** reports for teams across the company.
- Work with management, cloud, cybersecurity, IT and sales teams, and join client meetings.

Software Developer Intern · Babel — Nov 2025 – May 2026

- End-to-end modernisation of a full business management system: relational data model and business logic on **Oracle APEX** and **OutSystems ODC**.
- Migrated a legacy architecture to low-code, automated business processes and built AI proofs of concept.

PROJECTS

AquaAndalucía · AI forecasting of reservoir levels — 2025 – 2026

Bachelor's thesis · University of Málaga

- Level of **7 reservoirs** in Málaga from daily data, 2000–2024 (SAIH Hidrosur + AEMET), with feature engineering.
- Benchmarked **11 models** across 3 families (ARIMA/SARIMA/Prophet, Random Forest/XGBoost, and deep learning with 4 LSTMs and a graph-based GCN-LSTM), trained on the **Picasso** supercomputer (UMA).
- Multi-horizon evaluation against a persistence baseline: at 30 days the multivariate LSTM **halves the error** (RMSE 5.7 vs 12.0 pp) and at 60 days holds NSE at 0.70 while the baseline drops to 0.07.
- Full system: REST API (**FastAPI** + PyTorch) and a **React/TypeScript** web app with a map, 90-day what-if scenario simulation and an LLM natural-language assistant.

LOOKIT · Spanish fashion aggregator — 2026

Personal product · live · lookit.es

- Unifies the catalogues of **24 Spanish brands** (9,000+ products) from their Shopify stores and sends shoppers to each official site; the catalogue is rebuilt every hour.
- Natural-language search: a custom parser, with an LLM (Gemini) as fallback only when a query isn't understood.
- Account-free alerts for drops, price cuts and restocks via push (PWA) or email, with encrypted data (AES-256-GCM). Next.js, TypeScript, Redis, Vercel.

Client websites — 2026

Freelance · design, build and deployment

- **luciagascon.es**: a painter's portfolio with a walkable 3D gallery (**three.js**) and a custom admin panel (Cloudflare Pages, Functions and KV).
- **wildcamper.es**: van conversion, sales and rentals, with catalogue, booking calendar, admin panel with auth and RLS (**Supabase**) and an offline PWA.

Also: DeTemporada: PWA of seasonal food by Spanish region, with 210 products and 451 geolocated local shops.

EDUCATION

BSc in Computer Engineering · Computing major · University of Málaga — 2021 – 2026

Erasmus+ · Computer Science · Gdańsk University of Technology (Poland) · taught in English — 2024 – 2025

SKILLS

AI & data	Python, PyTorch, scikit-learn, XGBoost, pandas, statsmodels, LLMs (OpenAI, Gemini), KNIME
Backend & APIs	FastAPI, Node.js, Express, REST, SQL (Oracle, PostgreSQL, SQLite), Supabase, Redis, Java, C++
Cloud & DevOps	Cloudflare (Pages, Workers, KV), Vercel, GitHub Actions, Git, HPC (Picasso)
Frontend	TypeScript, React, Next.js, Angular, Tailwind CSS, three.js, PWA
Automation	Power Automate, Power BI, SharePoint, Oracle APEX, OutSystems ODC
Languages	Spanish (native) · English (C1)
Volunteering	Erasmus Student Network (2026 – present) · Fundación Madre Coraje (2025)